

The Positive Schur Property Does Not Characterize Universal WMP^+ Codomains

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Status. Candidate counterexample, likely valid, for human review.

1 Source question

Luiz and Miranda introduce the positive weak maximizing property: a pair of Banach lattices (E, F) has WMP^+ when every positive operator $T : E \rightarrow F$ with a positive non-weakly-null maximizing sequence is norm-attaining. After recalling that a Banach space Y has the Schur property exactly when (X, Y) has the ordinary WMP for every reflexive X , they ask whether there is a similar characterization for WMP^+ using the positive Schur property [1, Section 3, PDF p. 12].

It was proven by S. Dantas, M. Jung and G. Martínez-Cervantes that a Banach space Y has the Schur property if and only if the pair (X, Y) has the WMP for every reflexive space X (see [15] Theorem 3.5). It is a natural question to seek if there exists a similar characterization for the WMP^+ considering the positive Schur property instead of the Schur property.

Recall that a Banach lattice E has the positive Schur property if positive (or, equivalently, disjoint or positive disjoint) weakly null sequences in E are norm null. This property was introduced by W. Wnuk [31, 32] and F. Rübiger [28], and has been extensively studied for many experts, recent developments can be found, e.g., in [4, 6, 7, 8, 9, 29]. In the next, we give a sufficient condition so that the pair (E, F) has the WMP^+ whenever F has the positive Schur property.

Theorem 3.6. *If F has the positive Schur property, then the pair (E, F) satisfies the WMP^+ for every Banach lattice E such that B_E is sequentially absolutely weakly compact.*

The direct candidate characterization is:

F has the positive Schur property if and only if (E, F) has WMP^+ for every reflexive Banach lattice E .

The source proves the proposed sufficiency only when the unit ball of E is sequentially absolutely weakly compact, a condition stronger than reflexivity, and proves the relevant necessity result for order-continuous codomains. The construction below disproves sufficiency for reflexive domains.

2 Claimed contribution

Theorem 1. *For every $\delta > 0$ there is an order-continuous lattice norm $\|\cdot\|$ on $L_1[0, 1]$ such that*

$$\|h\|_1 \leq \|h\| \leq (1 + \delta)\|h\|_1 \quad (h \in L_1[0, 1]),$$

and the resulting Banach lattice $F = (L_1[0, 1], \|\cdot\|)$ has the positive Schur property, but

$$(L_2[0, 1], F)$$

fails WMP^+ . Consequently the positive Schur property does not characterize the codomains F for which (E, F) has WMP^+ for every reflexive Banach lattice E .

Thus the failure persists under arbitrarily small equivalent lattice renormings of the standard L_1 norm.

3 Proof intuition

An equivalent lattice renorming preserves the positive Schur property, but it can encode a moving family of positive functionals whose norms increase to an unattained limit. Rademacher half-sets provide exactly such a family. Their oscillatory parts vanish weakly against each fixed L_2 vector, preventing any single vector from attaining the limiting operator norm. On the other hand, each density nearly norms itself, so its normalization gives a maximizing sequence. Every density also has a fixed positive constant part; therefore that maximizing sequence converges weakly to a nonzero constant and is not weakly null. This is precisely the witness that violates WMP^+ .

4 Construction and proof

Work over the real scalars. Let (r_n) be the Rademacher functions on $[0, 1]$, normalized so that r_n takes the values -1 and 1 on sets of measure $1/2$. Put

$$A_n = \{t : r_n(t) = 1\}, \quad \mathbf{1}_{A_n} = \frac{\mathbf{1} + r_n}{2}.$$

Fix c with $0 < c \leq \delta$, choose any strictly increasing sequence $0 < c_n < c$ with $c_n \rightarrow c$ (for instance $c_n = cn/(n+1)$), and define

$$g_n = \mathbf{1} + c_n \mathbf{1}_{A_n}.$$

For $h \in L_1[0, 1]$, set

$$\|h\| = \sup_{n \geq 1} \int_0^1 g_n |h|. \tag{1}$$

Lemma 1. Equation (1) defines an order-continuous Banach lattice norm on $L_1[0, 1]$ satisfying

$$\|h\|_1 \leq \|h\| \leq (1 + c)\|h\|_1.$$

The resulting Banach lattice F has the positive Schur property.

Proof. Since $\mathbf{1} \leq g_n \leq (1 + c)\mathbf{1}$, the displayed norm equivalence is immediate. Positivity of the g_n , the inequality $|h + k| \leq |h| + |k|$, and monotonicity of the integral prove the norm and lattice-norm axioms. Completeness follows from equivalence with the ordinary L_1 norm. Since the ordinary L_1 norm is order continuous, the same equivalence shows that $\|\cdot\|$ is order continuous as well.

Equivalent norms have the same continuous linear functionals and hence the same weak topology. If $(h_j) \subset F_+$ is weakly null, then the constant functional $\mathbf{1} \in L_\infty$ gives

$$\|h_j\|_1 = \int_0^1 h_j \longrightarrow 0.$$

Consequently $\|h_j\| \leq (1 + c)\|h_j\|_1 \rightarrow 0$. This is the positive Schur property. $\square$

Consider the natural inclusion

$$T : L_2[0, 1] \longrightarrow F, \quad Tf = f.$$

It is positive. Write

$$d_n = \|g_n\|_2 = \left(1 + c_n + \frac{c_n^2}{2}\right)^{1/2}, \quad L = \left(1 + c + \frac{c^2}{2}\right)^{1/2}.$$

Then $d_n \rightarrow L$ and $d_n < L$ for every n .

Lemma 2. The operator T has norm L and does not attain its norm.

Proof. For $f \in L_2$, Cauchy–Schwarz gives

$$\|Tf\| = \sup_n \int_0^1 g_n |f| \leq \sup_n \|g_n\|_2 \|f\|_2 = L \|f\|_2.$$

For the reverse inequality let $x_n = g_n/d_n$. Then $x_n \geq 0$, $\|x_n\|_2 = 1$, and

$$\|Tx_n\| \geq \int_0^1 g_n x_n = d_n \longrightarrow L.$$

Thus $\|T\| = L$.

Fix now any $f \in L_2$ with $\|f\|_2 = 1$ and put $a_n = \int g_n |f|$. Since

$$g_n = \left(1 + \frac{c_n}{2}\right) \mathbf{1} + \frac{c_n}{2} r_n$$

and (r_n) is an orthonormal sequence in L_2 , its Fourier coefficients against the fixed vector $|f| \in L_2$ tend to zero. Hence

$$a_n \longrightarrow \left(1 + \frac{c}{2}\right) \|f\|_1 \leq 1 + \frac{c}{2}. \quad (2)$$

But

$$L^2 - \left(1 + \frac{c}{2}\right)^2 = \frac{c^2}{4} > 0,$$

so the limit in (2) is strictly less than L . Also, for each fixed n ,

$$a_n \leq \|g_n\|_2 \|f\|_2 = d_n < L.$$

Because (a_n) converges to a value below L , its tail is uniformly below L ; the finitely many remaining terms are each below L . Therefore

$$\|Tf\| = \sup_n a_n < L.$$

No unit vector f attains $\|T\| = L$. □

It remains to verify that the already constructed sequence is the witness required by WMP^+ . From

$$x_n = \frac{1 + c_n/2}{d_n} \mathbf{1} + \frac{c_n}{2d_n} r_n$$

and weak nullity of the orthonormal sequence (r_n) in L_2 , we obtain

$$x_n \xrightarrow{w} \frac{1 + c/2}{L} \mathbf{1} \neq 0.$$

Thus (x_n) is positive and non-weakly-null. The estimates in the proof of Lemma 2 show

$$d_n \leq \|Tx_n\| \leq \|T\| = L,$$

so (x_n) is a maximizing sequence for T . The positive operator T is not norm-attaining but admits a positive non-weakly-null maximizing sequence. Hence (L_2, F) fails WMP^+ , proving Theorem 1.

5 Verification and limitations

The separate adversarial report `verification.md` audits the Banach lattice norm, the positive Schur property, the exact operator norm, the strict nonattainment argument, and the maximizing sequence. No numerical or symbolic computation is used as proof. The only analytic ingredients are Cauchy–Schwarz, weak nullity of an orthonormal sequence, and elementary norm equivalence.

The construction answers the direct positive-Schur characterization negatively. It does not give an intrinsic replacement characterization of those F for which (E, F) has WMP^+ for every reflexive Banach lattice E .

6 Novelty check

The bounded search on 11 August 2026 covered the run’s registry, solution, attempt, and proof-gap indexes; exact searches for arXiv:2409.14625 and its title; the phrases “positive weak maximizing property” and “positive Schur property”; the published JMAA article; and arXiv searches for the exact question and close variants. The published 2026 version still presents the characterization as a natural question. No later proof or counterexample was found. This supports, but cannot certify, novelty.

References

- [1] J. L. P. Luiz and V. C. C. Miranda, *On norm-attaining positive operators between Banach lattices*, J. Math. Anal. Appl. 558 (2026), Paper No. 130379; arXiv:2409.14625, <https://arxiv.org/abs/2409.14625>.